

Root Cause Analysis Report

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Classification: Customer-facing technical report

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Status: Closed — corrective action verified

1. Executive Summary

A maritime LEO satellite communication terminal experienced intermittent link loss during sea trials. Field logs showed repeated modem re-registration events correlated with vessel motion and power rail dips. Root cause was identified as inadequate DC input filtering on the modem power feed combined with marginal RF cable grounding at the radome base. Corrective actions included installation of a marine-rated DC filter, rework of the ground strap, and a repeat SAT link-stability test. Post-correction testing achieved 48 hours continuous link with zero unplanned dropouts.

2. Issue Information

Field	Detail
Equipment	LEO Maritime VSAT Terminal (Model MTX-4500)
Customer / Vessel	[Anonymized] — Offshore support vessel
Reported date	2026-06-12
Severity	High — mission communication impact
Symptom	Intermittent link loss; modem re-registration every 15–45 min

3. Problem Description

The customer reported that the terminal established initial satellite lock successfully but lost connectivity repeatedly during vessel operation. Symptoms included:

- LED status cycling from LINK OK to SEARCHING
- NMS logs showing `MODEM\_REREG` events without corresponding RF obstruction

- Crew observation that events increased during heavy seas and main-engine load changes

No physical damage to the radome or antenna was observed at first inspection.

4. Environment

Parameter	Value
Location	Open sea, East China Sea trial area
Vessel motion	Roll $\pm 12^\circ$, pitch $\pm 8^\circ$ (moderate sea state)
Power source	Ship 24 VDC distribution panel, Branch CB-14
Ambient temperature	28–34 °C (enclosure internal 42 °C peak)
Installation age	6 weeks post-commissioning

5. Investigation & Test Evidence

5.1 Log review

Timestamp (UTC)	Event	Notes
2026-06-12 03:14	LINK_LOST	Preceded by `VIN_DIP 21.8V`
2026-06-12 03:14:02	MODEM_REREG	Recovery in 38 s
2026-06-12 05:41	LINK_LOST	`VIN_DIP 22.1V`, roll 11°
2026-06-12 08:22	LINK_LOST	No obstruction alarm

5.2 On-site measurements

Test	Result	Limit / Expectation
DC input at modem (steady state)	24.2 V	22–30 V
DC input during engine load step	21.6 V (200 ms dip)	>22 V recommended
Radome base ground resistance	4.8 Ω	<1 Ω per install guide
RF cable continuity	Pass	—
Blockage / obstruction	No active alarms	—

sensor		
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5.3 Failure symptom classification

- Category: Power integrity + grounding (installation-related)

- Not primary: Antenna pointing, modem firmware defect, or satellite network outage

6. Root Cause Analysis

6.1 Root cause (confirmed)

Inadequate DC input filtering on the modem power feed allowed transient undervoltage during ship load steps, triggering modem reset and link loss.

Contributing factor: Marginal RF ground strap at radome base ($4.8\ \Omega$) increased susceptibility to noise during motion, though primary trigger was power dip.

6.2 Why it was not caught at FAT

- Factory FAT used stable bench power supply; vessel distribution transients were not simulated.

- Ground strap torque was not verified with ohmmeter in SAT checklist (visual check only).

6.3 5-Why summary

1. Why link loss? → Modem re-registered repeatedly.

2. Why re-register? → Power or RF instability.

3. Why power instability? → 21.6-21.8 V dips on load step.

4. Why dips? → No marine DC filter; long cable run from panel.

5. Why missed? → SAT checklist did not include load-step power test.

7. Corrective Actions

ID	Action	Owner	Target date	Status
CA-01	Install marine-rated DC filter (Model DF-24M) at modem input	Field service	2026-06-18	Complete
CA-02	Replace and torque ground strap; verify $<1\ \Omega$	Field service	2026-06-18	Complete

CA-03	Update SAT checklist: power dip test under load step	QA / docs	2026-06-25	Complete
CA-04	Customer advisory bulletin for similar installations	Engineering	2026-07-01	Complete

8. Verification

Test	Procedure	Result	Pass/Fail
V-01	48 h continuous link test at sea	0 unplanned dropouts	**Pass**
V-02	Power dip test with engine load step	Min 23.1 V at modem	**Pass**
V-03	Ground resistance at radome base	0.6 Ω	**Pass**
V-04	NMS log review post-fix	No MODEM_REREG events	**Pass**

Verification sign-off: Field test lead, 2026-06-22.

9. Preventive Recommendations

1. Add load-step DC holdup test to all maritime SAT procedures.

2. Specify maximum ground resistance in installation work instructions.

3. Include power quality snapshot in commissioning evidence pack.

10. Attachments

- A: NMS event log excerpt (2026-06-12)

- B: Before/after ground strap photos

- C: SAT re-test record (48 h link stability)

This sample is anonymized for portfolio use. Structure aligns with Service 1 deliverable templates in `freelance/templates/`.